



# Fourier Convolution Block with global receptive field for MRI reconstruction

Haozhong Sun<sup>a</sup>, Yuze Li<sup>a</sup>, Zhongsen Li<sup>a</sup>, Runyu Yang<sup>a</sup>, Ziming Xu<sup>a</sup>, Jiaqi Dou<sup>a</sup>, Haikun Qi<sup>b</sup>, Huijun Chen<sup>a,\*</sup>

<sup>a</sup> Department of Biomedical Engineering, Tsinghua University, Beijing, China

<sup>b</sup> School of Biomedical Engineering, ShanghaiTech University, Shanghai, China

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## ABSTRACT

Reconstructing images from under-sampled Magnetic Resonance Imaging (MRI) signals significantly reduces scan time and improves clinical practice. However, Convolutional Neural Network (CNN)-based methods, while demonstrating great performance in MRI reconstruction, may face limitations due to their restricted receptive field (RF), hindering the capture of global features. This is particularly crucial for reconstruction, as aliasing artifacts are distributed globally. Recent advancements in Vision Transformers have further emphasized the significance of a large RF. In this study, we proposed a novel global Fourier Convolution Block (FCB) with whole image RF and low computational complexity by transforming the regular spatial domain convolutions into frequency domain. Visualizations of the effective RF and trained kernels demonstrated that FCB improves the RF of reconstruction models in practice. The proposed FCB was evaluated on four popular CNN architectures using brain and knee MRI datasets. Models with FCB achieved superior PSNR and SSIM than baseline models and exhibited more details and texture recovery. The code is publicly available at <https://github.com/Haozhong/FCB>.

## 1. Introduction

Magnetic resonance imaging (MRI) is a widely employed imaging modality for clinical diagnosis. However, the clinical application of MRI is impeded by its slow acquisition speed caused prolonged scan time and possible motion artifacts. Acquiring under-sampled data in the Fourier acquisition space, known as k-space, is a common approach to shorten the MRI scan time (Ravishanker and Bresler, 2010). However, acquiring fewer data points results in a violation of the Nyquist sampling criterion, leading to inevitable aliasing artifacts in the reconstructed images (Shannon, 1948).

Removing aliasing artifacts in under-sampled MRI scans presents a challenging inverse problem that requires incorporation of additional prior information. Compressed sensing (CS) technique (Lustig et al., 2007) utilizes the sparsity prior in predefined transforms and iterative optimization algorithms to address aliasing. However, CS requires manual selection of sparsity types and other parameters, and it often exhibits poor performance at higher under-sampling rates (Zbontar et al., 2018).

In recent years, deep learning (DL) based models have shown remarkable performance in MRI reconstruction. These models take advantages of capabilities of deep neural networks to recover images and exploit the prior knowledge obtained from the learning process

of data, surpassing traditional CS algorithms in terms of reconstruction quality and speed. There are two main approaches to utilizing DL models in MRI reconstruction. The first approach utilizes end-to-end models (Ronneberger et al., 2015; Lee et al., 2018; Han et al., 2019; El-Rewaidy et al., 2020), which are trained in a paired manner using under-sampled images and corresponding fully-sampled images without relying on other physical priors of MRI. The second approach involves unrolling existing iterative algorithms for reconstruction and replacing part of the iterations with neural networks (Sun et al., 2016; Zhang and Ghanem, 2018; Xiang et al., 2021; Hammernik et al., 2018; Schlemper et al., 2017; Sandino et al., 2020; Aggarwal et al., 2018). In this way, complex parameter selection of iterative algorithms can be optimized by learning, and the neural network serves as a more powerful regularization term.

In the realm of CS and DL based MRI reconstruction, undersampling patterns are typically random or uniform (Sriram et al., 2020b), rather than truncated as in super-resolution tasks (Chen et al., 2018). The resulting aliasing artifacts exhibit a long-range distribution in the image domain, requiring reconstruction models with a large receptive field (RF) to effectively remove these artifacts. However, most DL based MRI reconstruction models are based on Convolutional Neural Networks (CNNs), which have limited RF because of the local spatial convolutions

\* Corresponding author.

E-mail address: [chenhj\\_cbir@mail.tsinghua.edu.cn](mailto:chenhj_cbir@mail.tsinghua.edu.cn) (H. Chen).

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(e.g.,  $3 \times 3$  or  $5 \times 5$  kernels) used in most situations. Although deep networks with enough stacked convolution layers could cover large RF theoretically (Jaderberg et al., 2015), the effective receptive field (ERF) of CNNs falls significantly short of expectations (Ding et al., 2022). Recently, Vision Transformers (ViTs) have emerged as a powerful alternative (Dosovitskiy et al., 2020; Touvron et al., 2021; Liu et al., 2021), and outperform traditional CNNs by with stacked  $3 \times 3$  convolutions utilizing multi-head self-attention to capture global information. Other approaches have explored CNNs with larger kernels (Liu et al., 2022b; Ding et al., 2022; Liu et al., 2022a; Deng et al., 2023), achieving comparable, even superior, performance to ViTs. The increased ERF is considered a key factor in the success of these methods. However, both large kernel CNNs and ViTs face challenges such as extensive computational requirements and training difficulties (Liu et al., 2022a).

In this study, we propose the Fourier Convolution Block (FCB), a Plug-and-Play global operator that can be used in CNNs for MRI reconstruction. The term ‘Plug-and-Play’ used here is for indicating that FCB is a standalone convolution block, distinct from the pretrained model-based PnP iteration algorithm (Ahmad et al., 2020). FCB transforms standard spatial domain convolutions into the frequency domain to provide an extensive RF, leveraging the equivalence of spatial discrete domain convolution to element-wise multiplication in the frequency domain. FCB achieves a global RF, which matches the image size and can be adaptively learned, while needing less computational requirement than large kernel convolutions. To reduce the parameters of FCB, the regular convolutions are modified to depth separable convolutions (Howard et al., 2017), inspired by the design of Sun et al. (2023). This modification enables the Fourier Convolution’s application in the modern CNN with deep architecture. To further overcome the challenge in train networks with large RF (Liu et al., 2022a), a novel re-parameterization strategy for training networks with FCB was also proposed for better capturing both local and global information.

Thus, the main contributions of this work are:

1. Focusing on scaling up the RF of CNNs for MRI reconstruction, a Plug-and-Play Fourier operator FCB was proposed to replace local spatial convolutions. The results demonstrated that the proposed FCB could enlarge the ERF of CNNs and provide better reconstruction quality than ViTs and large-kernel CNNs with large RFs while costing less computing time.
2. We integrated FCB into a depth separable structure and proposed a novel re-parameterization strategy for training, enabling the network better fuse the local and global information. The two improvements pave the way for the efficient training of Fourier Convolution within a deep architecture.

## 2. Related work

### 2.1. DL-based MRI reconstruction

MRI reconstruction is a specific image recovery task, and DL-based models have been widely used for this purpose. End-to-End Model learns a direct mapping from aliased images to fully sampled images. UNet (Ronneberger et al., 2015), with its multi-scale structure and skipping connections, is one of the most popular architectures for MRI reconstruction and has been shown to outperform CS methods (Zbontar et al., 2018) at different acceleration ratios. Some studies have used separate networks for magnitude and phase reconstruction, with the dual-frame U-Net designed to avoid high-frequency signal loss (Lee et al., 2018). Complex-valued Networks have also been explored based on the complex nature of MRI signal (El-Rewaady et al., 2020; Cole et al., 2021; Duan and Ren, 2024). Besides, some works have incorporated perceptual loss or generative adversarial network (GAN) framework to improve the image quality of the reconstructed MRI (Yang et al., 2017; Mardani et al., 2018; Quan et al., 2018).

Taking the physical priors of MRI signals into consideration, the model-based network is designed as an unrolled form of the various

optimization algorithms for reconstruction. ADMNet (Sun et al., 2016) unrolled the alternating direction method of multipliers (ADMM) optimization in the general CS-based MRI model and represented different stages with learnable nodes, which can be trained using back-propagation. There are also other unrolled networks for the ISTA (Zhang and Ghanem, 2018) and FISTA (Xiang et al., 2021) optimization frameworks. VN (Variational Network) (Hammernik et al., 2018) formulated CS optimization as a variational model and learned parameters including the kernel, activation function, and data term. Additionally, other works used DL models as a regularization term, and the reconstruction iterations can be unrolled as a joint network and data consistency layer, realized by direct replacement (Schlemper et al., 2017; Duan et al., 2019; Yan et al., 2023), Proximal Gradient (Sandino et al., 2020), or Conjugate Gradient (CG) algorithm (Aggarwal et al., 2018).

### 2.2. ViT and large-kernel CNN

Recently, Vision Transformers have been shown to outperform CNNs in various visual tasks (Dosovitskiy et al., 2020; Touvron et al., 2021; Liu et al., 2021). ViT (Dosovitskiy et al., 2020) split the input image into patches and embeds them into vectors with fixed scale. Multi-head self-attention was then applied to establish long-range dependencies within the image. Swin Transformer (Liu et al., 2021) proposed a hierarchical structure similar to CNNs with shifted window-based self-attention, and has shown significant performance gains over previous state-of-the-art CNNs in various vision tasks. Recent studies (Lin and Heckel, 2022; Huang et al., 2022; Zhao et al., 2023; Guo et al., 2023) have also utilized ViTs for MRI reconstruction, leveraging the long-range dependence of ViTs to improve the reconstruction performance.

Inspired by the long-range dependence in ViTs, CNN designs with large kernel sizes have been explored, showing promising results in achieving comparable performance with ViTs (Liu et al., 2022b; Ding et al., 2022; Liu et al., 2022a; Deng et al., 2023). For instance, the kernel size in ConvNext (Liu et al., 2022b) was increased from  $3 \times 3$  to  $7 \times 7$  to match the size of the local window in Swin Transformer and Yan et al. (2024) proposed a ConvNeXt-based network for self-supervised MC MRI reconstruction. Further, Ding et al. (2022) scaled the normal convolution to  $31 \times 31$  using large depth-wise convolution, achieving comparable or better performance than Swin Transformer. Multi-branch convolutions with different kernel sizes (Liu et al., 2022a) were introduced and fused using softmax attention to adaptively adjust the receptive field size. Furthermore, Deng et al. (2023) shifted the sampling locations of convolutions in different channels to capture a global RF, and demonstrated improved results in semantic segmentation tasks.

### 2.3. Fourier convolution

Fourier convolution was initially proposed for accelerating convolutional operations due to its runtime efficiency (Mathieu et al., 2013). Spectral representation of CNNs using Fourier convolution was introduced in Rippel et al. (2015) and Ayat et al. (2019), where Fourier convolution serves as the spectral parametrization of convolutional layers. However, due to the large number of parameters in the Fourier convolution kernel, networks employing this approach lacked the depth seen in modern CNN architectures. To address this limitation, Wood and Larson (2021) parameterized the Fourier convolution kernel using given functions, allowing for a smaller number of trainable parameters to represent a Fourier convolution layer. However, this parametric approach still incurred high computational memory costs during forward and backward computations. We initially employ Fourier convolution within MRI reconstruction networks to capture large-scale correlations in aliased MR images. In our proposed approach, we integrate FCB with depth-wise convolution (Howard et al., 2017), significantly minimizing parameters and memory requirements. Furthermore, our re-parameterization method helps improve the convergence of Fourier convolution. As we know, our work represents the first instance of utilizing Fourier convolution in deep CNNs rather than in toy models.

### 3. Methods

#### 3.1. Fourier convolution

We will demonstrate the capability of Fourier Convolution to achieve both local and global RF in discrete scenarios. Fourier Convolution involves element-wise multiplication in the frequency domain. The Discrete Fourier Transform (DFT) transforms 2D images in the neural network to the frequency domain and is formulated as:

$$X(k_1, k_2) = \sum_{n=0}^{N-1} \sum_{m=0}^{N-1} \frac{x(n, m)}{N^2} \exp \left[ \frac{-2\pi j}{N} (nk_1 + mk_2) \right] \quad (1)$$

where  $x$  is the feature map with size of  $N \times N$ , and  $X$  is its spectrum. The Fourier Convolution is then formulated as the element-wise multiplication between the spectrum and a production kernel  $W$  with the same size:

$$Y(k_1, k_2) = X(k_1, k_2) \odot W(k_1, k_2) \quad (2)$$

With the definition of inverse DFT (IDFT):

$$x(n, m) = \sum_{k_1=0}^{N-1} \sum_{k_2=0}^{N-1} X(k_1, k_2) \exp \left[ \frac{2\pi j}{N} (nk_1 + mk_2) \right] \quad (3)$$

it could be deduced that there is an equivalence between Fourier Convolution and spatial convolution:

$$\text{IDFT}\{Y\} = x * w \quad (4)$$

where “ $*$ ” represents convolution with circle-padding:

$$x * w = \sum_{p=0}^{N-1} \sum_{q=0}^{N-1} x(p, q) w(\lfloor (n-p)/N \rfloor, \lfloor (m-q)/N \rfloor) \quad (5)$$

The convolution kernel  $w$  is the IDFT of FCB kernel  $W$  with size  $N \times N$  but it can be expressed as a zero-padding version of some smaller kernel  $w'$  with a size of  $K(1 \leq K \leq N)$ :

$$w(n, m) = \begin{cases} w'(n, m), & n \leq K, m \leq K \\ 0, & \text{others} \end{cases} \quad (6)$$

It implies that, although production kernel  $W$  in Fourier domain has a constant size as the same as the input, its spatial counterpart could be a zero-padded convolution kernel of which the size  $K \times K$  varies from  $1 \times 1$  to the input size  $N \times N$ . This point highlights that Fourier convolution can result in a global RF, as it could be equivalent to a convolution with a kernel size that matches the input image. The local RF is also accessible when the Fourier convolution corresponds the spatial one with small size.

#### 3.2. Fourier Convolution Block (FCB) for CNNs

In this study, a Plug-and-Play Fourier Convolution Block (FCB) which can replace the traditional convolution layers of popular CNNs for MRI reconstruction is proposed. The illustration of FCB is shown in Fig. 1. The input image  $x \in \mathbb{R}^{N \times N}$  is firstly transformed to frequency domain using Fast Fourier Transform (FFT), denoted as  $F$ , and element-wise multiplied by the learned kernel  $W \in \mathbb{C}^{N \times N}$  in the frequency domain. The resulting spectrum is then transformed back to the spatial domain using Inverse FFT (IFFT), denoted as  $F^{-1}$ . This process can be formulated as

$$y = F^{-1}\{F\{x\} \otimes W\} \quad (7)$$

$\otimes$  represents the element-wise multiplication. Because the FFT transforms the real-valued  $x$  into a complex-valued spectrum, this multiplication is a complex operation. Using conjugate symmetry similar to Rippel et al. (2015), half of the frequency data is stored and operated on. The result of the multiplication can be transformed back to a real value using the IFFT.

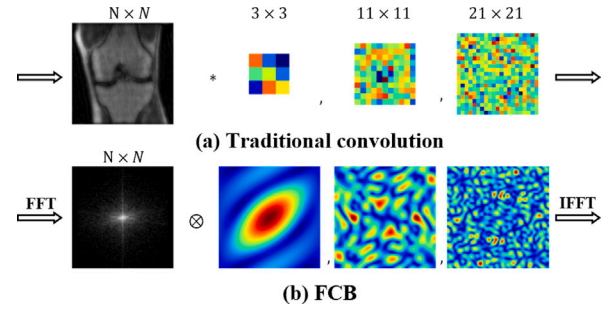


Fig. 1. The illustration of the proposed Fourier Convolution Block (FCB). (a) Three traditional convolution layers with kernel sizes of  $3 \times 3$ ,  $11 \times 11$ , and  $21 \times 21$ . (b) The equivalent FCBs with the size of  $N \times N$  corresponding to the traditional convolution layers with different kernel size shown on the top.

Table 1

Comparison of the computation operations and parameters between regular convolution and FCB.

|            | Convolution                              | FCB                                         |
|------------|------------------------------------------|---------------------------------------------|
| Operations | $S \cdot f \cdot f' \cdot N^2 \cdot K^2$ | $S \cdot f \cdot N^2 \cdot (f' + 4 \log N)$ |
| Parameters | $f \cdot f' \cdot K^2$                   | $f \cdot f' \cdot K^2$                      |

According to the conclusion of Eq. (4), Fourier Convolution can strictly correspond to any spatial convolution with arbitrary size between  $1 \times 1$  to  $N \times N$  and FCB is capable of covering both local and global RFs. All operations in FCB are continuously differentiable, enabling the adaptive learning of the effective RFs for each FCB layer through backpropagation during training.

The theoretical computation operations and parameters of FCB and traditional convolution are analyzed and shown in Table 1. With the image size  $N \times N$ , convolution kernel size  $K \times K$ , batch size  $S$ , the number of input channels  $f'$  and the number of output channels  $f$ , the number of operations for a traditional convolution should be  $S f f' N^2 K^2$ . On the other hand, the proposed FCB includes a relatively simple element-wise product with  $f' N^2$  operations, FFT and IFFT with operations  $2N^2 \log N$ . Thus, the total number of operations of FCB is  $S f N^2 (f' + 4 \log N)$ . Notably, only half of the frequency data needs to be computed and stored in FCB since the frequency data satisfies conjugate symmetry.

The ratio of the two computational complexities is expressed as  $(f' + 4 \log N)/(f' K^2)$ , and in our study,  $f'$  is 1 due to the depth-wise convolution. This implies that FCB may have more operations than traditional convolution when the kernel size is small. However, to enlarge the RF, increasing the kernel size  $K$  will be more computationally expensive than using FCB, as demonstrated by the results in Section 4.6. Conversely, FCB incurs the substantial  $N^2$  parameters when convolution parameters depend on kernel size.

#### 3.3. FCB enhanced CNN

In this study, FCB was tested in four popular networks for MRI reconstruction: UNet (Ronneberger et al., 2015), E2EVar (Sriram et al., 2020a), MoDL (Aggarwal et al., 2018) and VSNet (Duan et al., 2019), representing one end-to-end CNN and two model-based CNN architectures, respectively. The original architectures of these models are shown in Fig. 2. To incorporate FCB, traditional convolution blocks were modified to depth separable convolutions (Howard et al., 2017) with identity shortcuts, in which the depth-wise (DW) can be replaced by FCB, as illustrated in Fig. 2. Since the network backbone in E2EVar is UNet, we did not separately illustrate its modification version.

Here, the depth separable convolution (Howard et al., 2017) was to address the disadvantage of the substantial parameters of FCB, since the majority of parameters in depth separable convolutions are

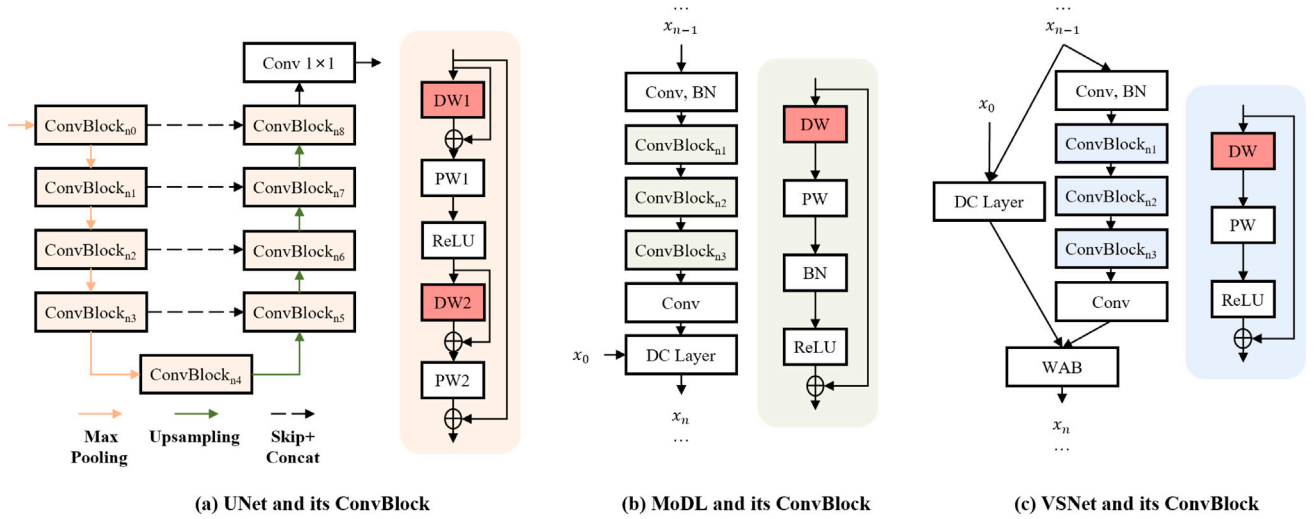


Fig. 2. The architecture of baseline models and proposed convolution blocks (only one iteration in MoDL and VSNet is shown). Abbrev: DW = Depth-wise Convolution, PW = Point-Wise Convolution. The convolution blocks are consistently colored with the model views, with DW layers highlighted in red. This indicates that the convolution operation here can be replaced by FCB.

concentrated in the  $1 \times 1$  convolution. Moreover, identity shortcut connections were also used since they are crucial in large-kernel models or transformers (Liu et al., 2022b).

CNN models with large kernel pose a challenge in terms of scaling to a large RF while retaining local information. To address this challenge, RepLKNet (Ding et al., 2022) proposed the idea of paralleling a small kernel convolution group and re-parameterizing the parallel branch into the main branch. Inspired by that, a re-parameterization training strategy was proposed to tackle this issue. We took the advantage of adaptive RF of FCB to synthesize information from different scales, and avoided extra parallels and complex optimization. In this training strategy, the original network with traditional spatial  $3 \times 3$  convolution kernels were firstly trained to capture local patterns. Once the first phase converges, the DW convolution with FCB was re-parameterized from  $3 \times 3$  convolution to Fourier convolution, then the network was fine-tuned to scale to a global RF. As per Eqs. (4) and (6), this re-parameterization is essentially a zero-padded DFT of the convolution kernel  $w \in \mathbb{R}^{K \times K}$ . The transformed FCB kernel can be formulated as:

$$W(k_1, k_2) = \frac{1}{N^2} \sum_{n=0}^{K-1} \sum_{m=0}^{K-1} w(n, m) \exp \left[ \frac{-2\pi j}{N} (k_1 n + k_2 m) \right] \quad (8)$$

To ensure a strict equivalence between FCB and local convolution in the first training phase, circular convolution was used, which also agrees with the naturally wrapped MRI signal (Sandino et al., 2020).

## 4. Experiments

### 4.1. Data preparation

In this study, brain and knee data of FastMRI dataset (Zbontar et al., 2018) were used to test the performance of the proposed FCB. For brain experiment, 250 volumes of T1, T2, T2 fluid-attenuated inversion recovery (FLAIR) and post-contrast T1 brain data (40, 50, 135, 25 respectively) were included. Among them, 200 volumes (3154 slices) were used for training and 50 (794 slices) for validation with the same composition rate of contrasts. For knee experiment, 118 volumes of PD weighted knee data were involved, and 100 (2963 slices) were used for training, 18 (524 slices) for validation. All the images were center-cropped to  $320 \times 320$ , and the maximum magnitude of each volume was normalized to 1. Because the number of coils of brain dataset was not identical, coil compression was performed by keeping the first  $n_c$  columns singular vectors of singular value decomposition (SVD) (Klema and Laub, 1980). In this study,  $n_c$  was set to 8. The sensitive maps were estimated from the center k-space region by using ESPIRiT (Roy and Kailath, 1989).

### 4.2. Baselines and evaluation strategies

The performance of UNet with the proposed FCB (F-UNet), E2EVar with FCB (F-E2EVar), MoDL with FCB (F-MoDL) and VSNet with FCB (F-VSNet), as well as baseline UNet (Ronneberger et al., 2015), E2EVar (Sriram et al., 2020a), MoDL (Aggarwal et al., 2018) and VSNet (Duan et al., 2019) models, were evaluated and compared. Additionally, three ViT models for MRI reconstruction were compared including ViT (Lin and Heckel, 2022), ReConFormer (Guo et al., 2023) and SwinMR (Huang et al., 2022). Methods utilizing k-space data including Ksp (Han et al., 2019) and MDRecon (Ran et al., 2020) for global dependency were also considered. Furthermore, the Fast Fourier Convolution (FFC) method (Chi et al., 2020) employed  $1 \times 1$  convolution and nonlinear layer in the Fourier domain to achieve a global receptive field. We compared the FasterFC block (Liu et al., 2024b,a), adapted from FFC for MRI reconstruction in the UNet architecture (FasterFC-UNet), with the FCB.

The performance of FCB was evaluated on the previously described brain and knee datasets at  $8\times$  and  $12\times$  acceleration rate, using a variable density Poisson under-sampling pattern (Lustig et al., 2008). The size of the Autocalibration Signal (ACS) was set to  $20 \times 20$ , and the polynomial order of variable density was set to 4. Additionally, the experiments were extended to include Cartesian and radial under-sampling patterns. Models' performance under extra noise was also assessed. Gaussian noise with a standard deviation equivalent to 10% and 20% of the signal intensity was introduced into the under-sampled k-space data. Peak Signal to Noise Ratio (PSNR) and Structural Similarity (SSIM) on the validation set were used for performance evaluation, computed on the magnitude image of the reconstruction results compared to the "ground truth" images reconstructed with fully-sampled k-space. The Wilcoxon signed-rank test was utilized to evaluate the improvements in the two metrics. To ensure a fair comparison between End-to-End and model-based methods, the ground truth images were combined using the sensitive maps from ESPIRiT (Roy and Kailath, 1989).

### 4.3. Implementation details

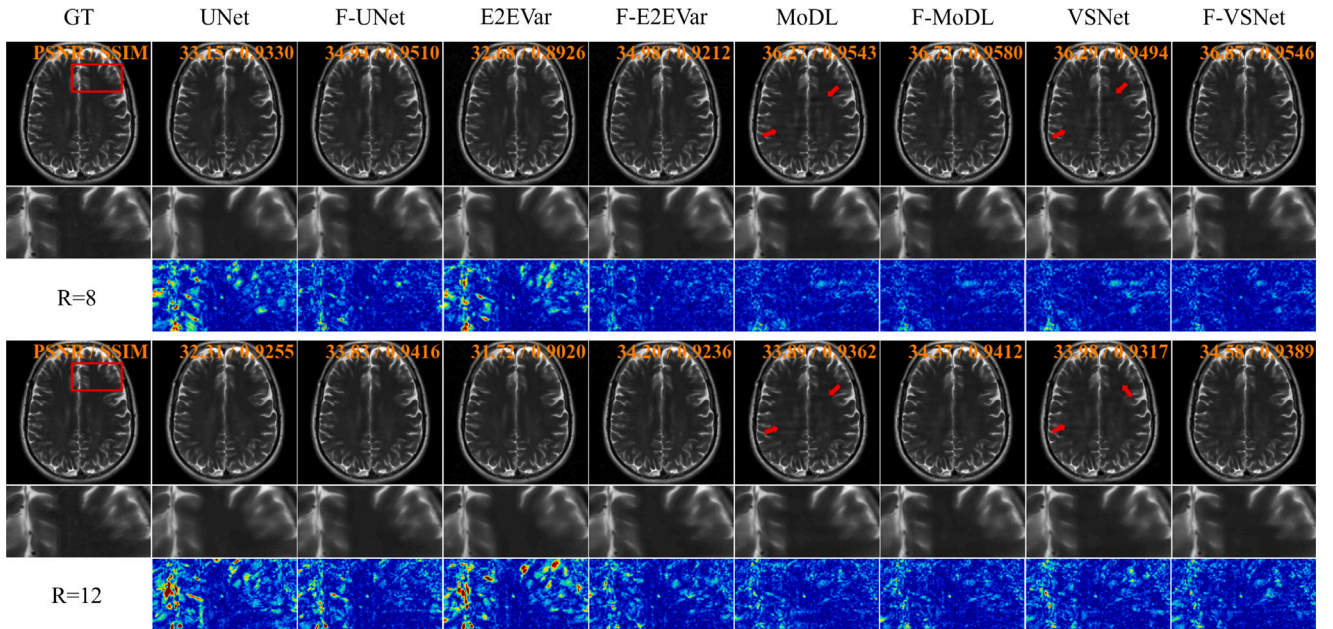
In all the models, mean absolute error (MAE) loss on the coil-combined image was utilized for training:

$$L(x, \hat{x}) = \frac{1}{S} \sum_{s=1}^S |x_s - \hat{x}_s| \quad (9)$$

**Table 2**

Quantitative results on the validation set in brain and knee datasets at 8× and 12× acceleration.

| Task  | Method   | PSNR         |              | SSIM            |                 |
|-------|----------|--------------|--------------|-----------------|-----------------|
|       |          | 8×           | 12×          | 8×              | 12×             |
| Brain | UNet     | 34.44 ± 3.38 | 33.19 ± 3.22 | 0.9047 ± 0.1130 | 0.8909 ± 0.1127 |
|       | F-UNet   | 35.25 ± 3.48 | 34.11 ± 3.34 | 0.9126 ± 0.1133 | 0.9013 ± 0.1133 |
|       | p        | ***          | ***          | ***             | ***             |
|       | E2EVar   | 34.22 ± 3.17 | 33.20 ± 3.15 | 0.8711 ± 0.1082 | 0.8771 ± 0.1109 |
|       | F-E2EVar | 35.71 ± 3.27 | 34.76 ± 3.16 | 0.8879 ± 0.1094 | 0.8931 ± 0.1122 |
|       | p        | ***          | ***          | ***             | ***             |
|       | MoDL     | 36.62 ± 3.27 | 35.41 ± 3.30 | 0.9165 ± 0.1125 | 0.9024 ± 0.1179 |
|       | F-MoDL   | 37.15 ± 3.43 | 35.63 ± 3.30 | 0.9192 ± 0.1129 | 0.9038 ± 0.1173 |
|       | p        | ***          | ***          | ***             | ***             |
|       | VSNet    | 36.78 ± 3.20 | 35.06 ± 2.87 | 0.9140 ± 0.1112 | 0.8982 ± 0.1110 |
|       | F-VSNet  | 37.14 ± 3.21 | 35.59 ± 2.98 | 0.9182 ± 0.1109 | 0.9068 ± 0.1120 |
|       | p        | ***          | ***          | ***             | ***             |
| Knee  | UNet     | 34.23 ± 2.15 | 33.09 ± 2.06 | 0.8955 ± 0.0338 | 0.8731 ± 0.0382 |
|       | F-UNet   | 35.52 ± 2.34 | 34.36 ± 2.22 | 0.9136 ± 0.0321 | 0.8943 ± 0.0355 |
|       | p        | ***          | ***          | ***             | ***             |
|       | E2EVar   | 33.59 ± 2.13 | 32.67 ± 2.04 | 0.8748 ± 0.0346 | 0.8617 ± 0.0362 |
|       | F-E2EVar | 35.32 ± 2.42 | 34.48 ± 2.30 | 0.9025 ± 0.0342 | 0.8951 ± 0.0346 |
|       | p        | ***          | ***          | ***             | ***             |
|       | MoDL     | 35.04 ± 2.43 | 33.66 ± 2.39 | 0.8974 ± 0.0374 | 0.8717 ± 0.0437 |
|       | F-MoDL   | 35.68 ± 2.56 | 34.38 ± 2.37 | 0.9063 ± 0.0375 | 0.8851 ± 0.0409 |
|       | p        | ***          | ***          | ***             | ***             |
|       | VSNet    | 35.31 ± 2.30 | 33.72 ± 2.14 | 0.9030 ± 0.0332 | 0.8719 ± 0.0401 |
|       | F-VSNet  | 35.84 ± 2.41 | 34.37 ± 2.25 | 0.9111 ± 0.0327 | 0.8869 ± 0.0367 |
|       | p        | ***          | ***          | ***             | ***             |

\*\*\* means that  $p < 0.001$ .

**Fig. 3.** A reconstruction example of the T2-weighted data in validation set in brain dataset. The top displays the result at 8× acceleration, while the bottom displays the result at 12× acceleration. PSNR and SSIM of the single image reconstructed are noted in the top right-hand corner. The second and fifth rows depict the reconstruction of the zoomed region marked in the ground truth. The third and sixth rows display the residual error in this zoomed region, with all errors multiplied by 10 for better visualization.

where  $x_s$  is the  $s$ th target image and  $\hat{x}_s$  is the predicted one.  $S$  is the size of mini-batch, which is 16 for UNet and VSNet and 8 for MoDL. The baseline UNet had 64 channels in the hidden layer and involved 4 times down-sampling. The E2EVar model had a total of 12 iterations, with its UNet backbone consisting of 18 channels. The MoDL model included 10 iterations with 64 filters in each of the 5 convolution layers and the VSNet model had 5 iterations with the same width and depth.

Notably, in the four baseline models, the convolution blocks were replaced with the depth-separable convolutions with shortcut and the padding mode was circular padding to ensure a fair comparison. The

last 6 layers in F-UNet were replaced by the proposed FCB, while both in F-MoDL and F-VSNet, the 3rd and 4th layers were replaced.

When training the UNet, F-UNet, and ViT models, all images were handled as magnitude images. For the E2EVar, MoDL, VSNet and their FCB versions, the real and imaginary values were arranged as separate channels (Aggarwal et al., 2018). The baseline models and their FCB versions were trained using the RMSprop optimizer (Graves, 2013) with a learning rate of  $1e-4$  for UNet and VSNet, and  $1e-3$  for MoDL. E2EVar was trained with the Adam optimizer (Kingma, 2014) at a  $1e-3$  learning rate, as in Sriram et al. (2020a). The baseline models were trained for

**Table 3**  
Quantitative results on the validation set using Cartesian and Radial mask.

| Task  | Method   | PSNR             |                  | SSIM                |                     |
|-------|----------|------------------|------------------|---------------------|---------------------|
|       |          | Cartesian        | Radial           | Cartesian           | Radial              |
| Brain | UNet     | 31.21 $\pm$ 3.02 | 34.36 $\pm$ 3.06 | 0.8736 $\pm$ 0.1109 | 0.9087 $\pm$ 0.1122 |
|       | F-UNet   | 31.74 $\pm$ 2.99 | 34.98 $\pm$ 3.12 | 0.8811 $\pm$ 0.1113 | 0.9150 $\pm$ 0.1133 |
|       | p        | ***              | ***              | ***                 | ***                 |
|       | E2EVar   | 31.57 $\pm$ 2.95 | 33.89 $\pm$ 3.02 | 0.8740 $\pm$ 0.1103 | 0.8682 $\pm$ 0.1079 |
|       | F-E2EVar | 32.54 $\pm$ 2.98 | 35.37 $\pm$ 3.13 | 0.8858 $\pm$ 0.1124 | 0.8956 $\pm$ 0.1109 |
|       | p        | ***              | ***              | ***                 | ***                 |
|       | MoDL     | 30.28 $\pm$ 3.11 | 35.61 $\pm$ 3.05 | 0.8426 $\pm$ 0.1160 | 0.9141 $\pm$ 0.1116 |
|       | F-MoDL   | 30.82 $\pm$ 3.03 | 35.98 $\pm$ 3.07 | 0.8519 $\pm$ 0.1153 | 0.9197 $\pm$ 0.1118 |
|       | p        | ***              | ***              | ***                 | ***                 |
|       | VSNet    | 29.91 $\pm$ 2.81 | 35.77 $\pm$ 2.94 | 0.8354 $\pm$ 0.1063 | 0.9184 $\pm$ 0.1104 |
|       | F-VSNet  | 30.46 $\pm$ 2.77 | 36.03 $\pm$ 2.95 | 0.8476 $\pm$ 0.1074 | 0.9208 $\pm$ 0.1105 |
|       | p        | ***              | ***              | ***                 | ***                 |
| Knee  | UNet     | 31.40 $\pm$ 1.94 | 34.06 $\pm$ 2.25 | 0.8456 $\pm$ 0.0386 | 0.8991 $\pm$ 0.0340 |
|       | F-UNet   | 31.94 $\pm$ 1.96 | 35.01 $\pm$ 2.38 | 0.8557 $\pm$ 0.0369 | 0.9112 $\pm$ 0.0332 |
|       | p        | ***              | ***              | ***                 | ***                 |
|       | E2EVar   | 31.12 $\pm$ 1.94 | 33.52 $\pm$ 2.19 | 0.8338 $\pm$ 0.0392 | 0.8756 $\pm$ 0.0347 |
|       | F-E2EVar | 32.17 $\pm$ 2.04 | 34.92 $\pm$ 2.43 | 0.8549 $\pm$ 0.0385 | 0.9012 $\pm$ 0.0349 |
|       | p        | ***              | ***              | ***                 | ***                 |
|       | MoDL     | 29.81 $\pm$ 2.16 | 35.15 $\pm$ 2.40 | 0.8046 $\pm$ 0.0429 | 0.8997 $\pm$ 0.0368 |
|       | F-MoDL   | 30.71 $\pm$ 2.08 | 35.50 $\pm$ 2.46 | 0.8231 $\pm$ 0.0431 | 0.9085 $\pm$ 0.0353 |
|       | p        | ***              | ***              | ***                 | ***                 |
|       | VSNet    | 29.86 $\pm$ 1.79 | 35.30 $\pm$ 2.38 | 0.7986 $\pm$ 0.0435 | 0.9042 $\pm$ 0.0367 |
|       | F-VSNet  | 30.48 $\pm$ 1.80 | 35.55 $\pm$ 2.46 | 0.8139 $\pm$ 0.0424 | 0.9080 $\pm$ 0.0367 |
|       | p        | ***              | ***              | ***                 | ***                 |

80 epochs, except for VSNet which was trained for 200 epochs. The FCB enhanced models were trained for the same number of epochs to the corresponding baseline models. For the training of F-UNet, F-E2EVar, F-MoDL and F-VSNet, after re-parameterization, the learning rate was set to  $1e-4$ , and the L2 weight normalization was added. The optimizer procedure for ViT (Lin and Heckel, 2022), ReconFormer (Guo et al., 2023), SwinMR (Huang et al., 2022), Ksp (Han et al., 2019) and MDRecon (Ran et al., 2020) were kept consistent with their original papers.

#### 4.4. Quantitative metrics results

The comparison between FCB and traditional convolution in four architectures (UNet, E2EVar, MoDL and VSNet) at  $8\times$  and  $12\times$  acceleration rates in brain and knee validation datasets are shown in Table 2. The mean value and standard deviation of PSNR and SSIM are included. Across different acceleration rates and datasets, FCB outperforms its baseline counterpart by a significant margin in all four architectures. The F-E2EVar demonstrates the highest performance improvement. For all architectures, the Wilcoxon signed-rank test results indicate statistical significance ( $p < 0.001$ ). The performance difference in brain and knee data could be attributed to part of reconstructions with inferior quality, characterized by more noticeable artifacts in knee images shown in Section 4.5.

Table 3 shows the results when equispaced Cartesian and radial under-sampling patterns were utilized at  $8\times$  acceleration. The proposed FCB significantly improved the performance of the four architectures across different patterns. The overall metrics of the four baseline models in the equispaced Cartesian undersampling were low, while the implementation of FCB led to remarkable improvements in all architectures. Then, Table 4 presents the quantitative results when additional noise was introduced. The inclusion of noise led to a decrease in the PSNR and SSIM values for the baseline models, particularly for MoDL. For all four architectures, FCB significantly outperformed its baseline counterpart, with F-VSNet achieving the best PSNR and SSIM at both 10% and 20% noise levels.

The comparison between F-MoDL and other methods including three ViT models (ViT, ReconFormer, SwinMR) and two k-space data

**Table 4**  
Quantitative results on the validation set when 10% Or 20% Gaussian noise added in brain dataset at  $8\times$  acceleration.

| Method   | PSNR                               |                                    | SSIM                                  |                                       |
|----------|------------------------------------|------------------------------------|---------------------------------------|---------------------------------------|
|          | 10%                                | 20%                                | 10%                                   | 20%                                   |
| UNet     | 34.00 $\pm$ 3.18                   | 33.35 $\pm$ 3.05                   | 0.9015 $\pm$ 0.1121                   | 0.8947 $\pm$ 0.1119                   |
| F-UNet   | 34.66 $\pm$ 3.23                   | 34.00 $\pm$ 3.07                   | 0.9088 $\pm$ 0.1129                   | 0.9035 $\pm$ 0.1122                   |
| p        | ***                                | ***                                | ***                                   | ***                                   |
| E2EVar   | 33.64 $\pm$ 2.98                   | 32.83 $\pm$ 2.78                   | 0.8596 $\pm$ 0.1058                   | 0.8329 $\pm$ 0.1015                   |
| F-E2EVar | 34.79 $\pm$ 3.01                   | 33.77 $\pm$ 2.76                   | 0.8749 $\pm$ 0.1071                   | 0.8469 $\pm$ 0.1025                   |
| p        | ***                                | ***                                | ***                                   | ***                                   |
| MoDL     | 35.55 $\pm$ 2.94                   | 34.21 $\pm$ 2.85                   | 0.9026 $\pm$ 0.1090                   | 0.8833 $\pm$ 0.1191                   |
| F-MoDL   | 35.96 $\pm$ 3.01                   | 34.54 $\pm$ 2.75                   | 0.9114 $\pm$ 0.1104                   | 0.8982 $\pm$ 0.1093                   |
| p        | ***                                | ***                                | ***                                   | ***                                   |
| VSNet    | 35.86 $\pm$ 2.87                   | 34.56 $\pm$ 2.64                   | 0.9112 $\pm$ 0.1102                   | 0.8942 $\pm$ 0.1090                   |
| F-VSNet  | <b>36.21 <math>\pm</math> 2.93</b> | <b>34.92 <math>\pm</math> 2.72</b> | <b>0.9172 <math>\pm</math> 0.1110</b> | <b>0.9057 <math>\pm</math> 0.1101</b> |
| p        | ***                                | ***                                | ***                                   | ***                                   |

involved models (Ksp, MDRecon) is shown in Table 5. The proposed FCB method achieved significantly better PSNR and SSIM than all these methods. Additionally, the parameters and runtime of the networks were compared. Runtime was measured by the mean GPU time of the forward pass on the validation dataset. The parameter count of F-MoDL was lower than most methods, except for MDRecon. Its runtime was shorter than the three ViT models with time-consuming self-attention, but longer than Ksp and MDRecon due to the conjugate gradient operator.

#### 4.5. Qualitative visualization results

Figs. 3 and 4 illustrate the reconstructed brain T2 and T1 contrast images from all models. UNet and E2EVar produced smoother results, while MoDL and VSNet exhibited visible aliasing artifacts (marked by red arrows in Fig. 3), possibly due to aliasing and texture confusion. On the other hand, F-UNet and F-E2EVar recovered more details than UNet and E2EVar, and both F-MoDL and F-VSNet effectively suppressed the aliasing artifacts presented in their baselines. The zoomed region

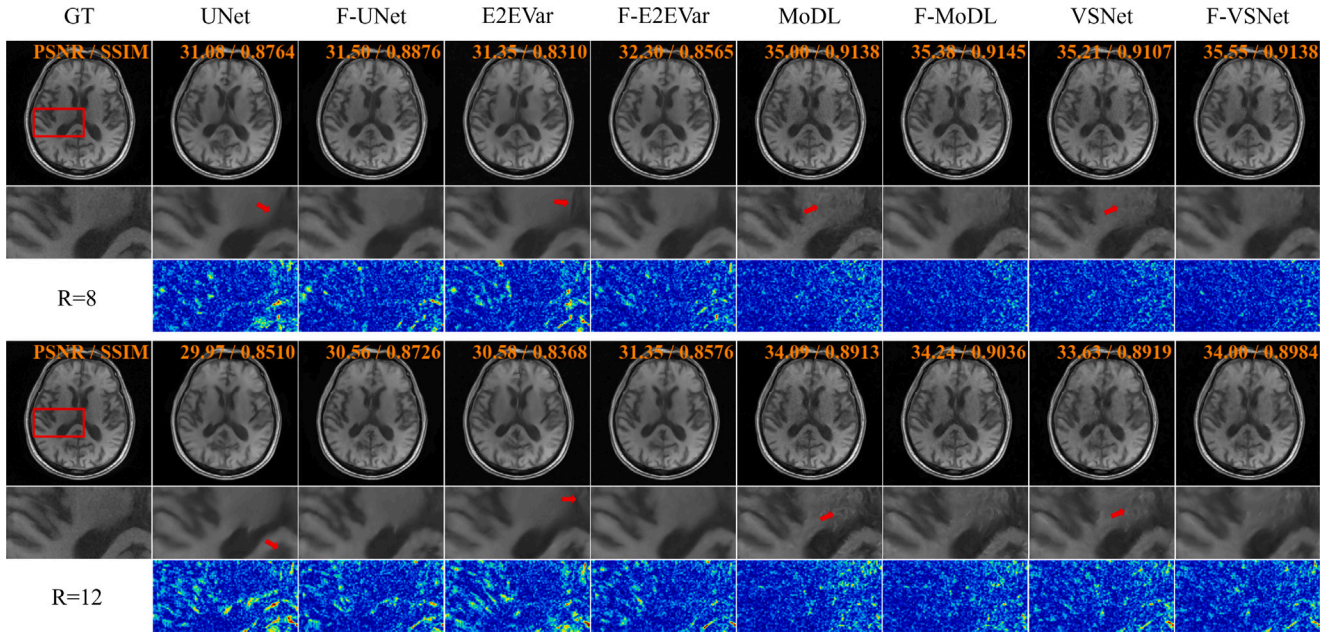


Fig. 4. A reconstruction example of the T1-weighted data in validation set in brain dataset.

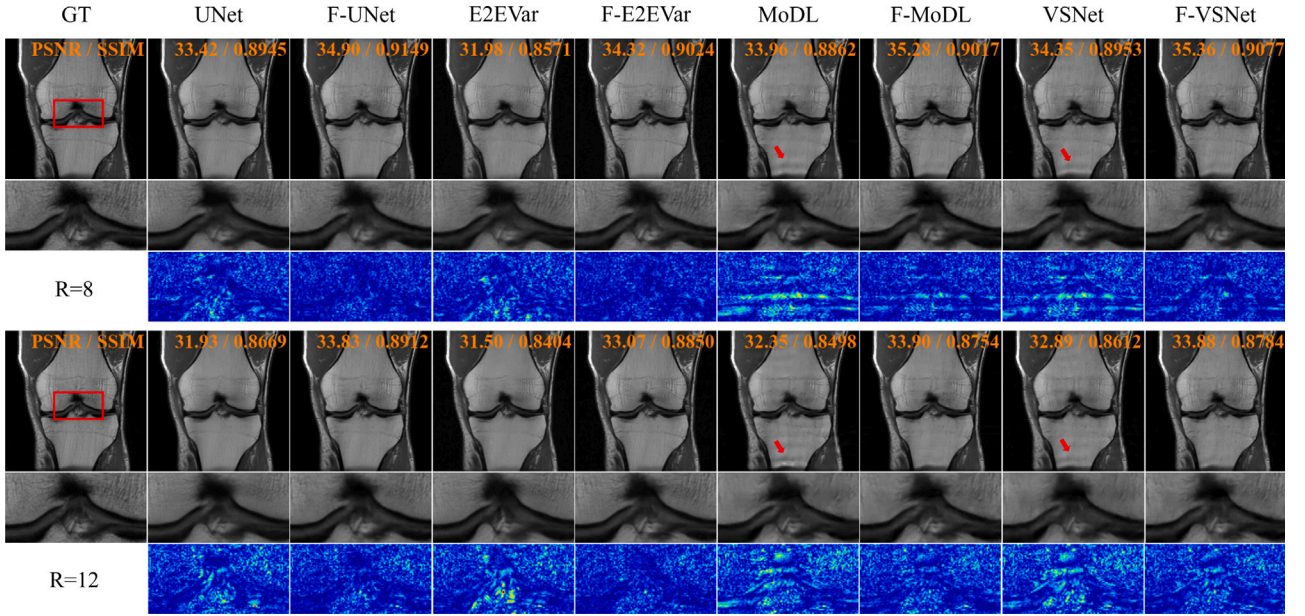


Fig. 5. A reconstruction example of the validation set in knee dataset. The top displays the result at 8× acceleration, while the bottom displays the result at 12× acceleration.

Table 5

Quantitative results compared with other methods.

|               | F-MoDL                 | ViT             | ReconFormer     | SwinMR          | Ksp             | MDRecon         |
|---------------|------------------------|-----------------|-----------------|-----------------|-----------------|-----------------|
| PSNR          | <b>37.15 ± 3.40</b>    | 31.98 ± 2.97    | 33.69 ± 3.04    | 34.94 ± 3.27    | 33.57 ± 3.51    | 36.31 ± 3.49    |
| p             | —                      | ***             | ***             | ***             | ***             | ***             |
| SSIM          | <b>0.9197 ± 0.1123</b> | 0.8828 ± 0.1123 | 0.8979 ± 0.1091 | 0.9069 ± 0.1122 | 0.8610 ± 0.1100 | 0.8930 ± 0.1164 |
| p             | —                      | ***             | ***             | ***             | ***             | ***             |
| Parameter (M) | 6.61                   | 616.71          | 1.08            | 9.64            | 30.34           | 1.3             |
| Time (ms)     | 50.1                   | 84.96           | 596.94          | 297.3           | 6.01            | 17.94           |

highlighted intricate details and sharp edges, where UNet and E2EVar exhibited more residual errors and MoDL and VSNet generated some false textures (marked by red arrows in Fig. 4(b)), compared with models with FCB. Overall, models with FCB significantly enhanced reconstruction accuracy, displaying less false structure and texture.

The knee images reconstructed by all models are shown in Fig. 5. At 8× acceleration, UNet produced smoother result, while MoDL and VSNet generated obvious horizontal aliasing artifacts (marked by red arrows), similar to the results of the brain dataset. By employing FCB, F-UNet and F-E2EVar recovered more details, especially the bone texture,

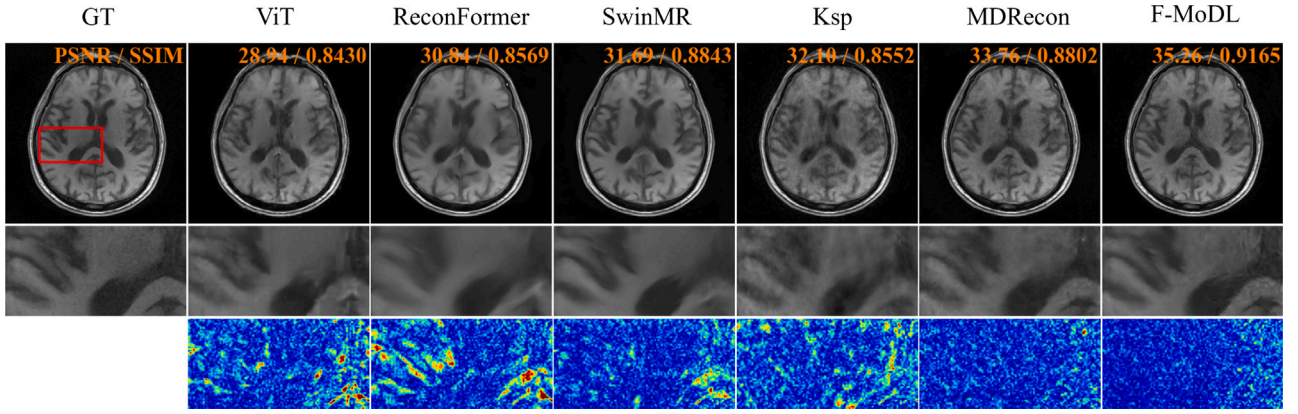


Fig. 6. An example of brain T1-weighted data reconstructed by F-MoDL and other methods.

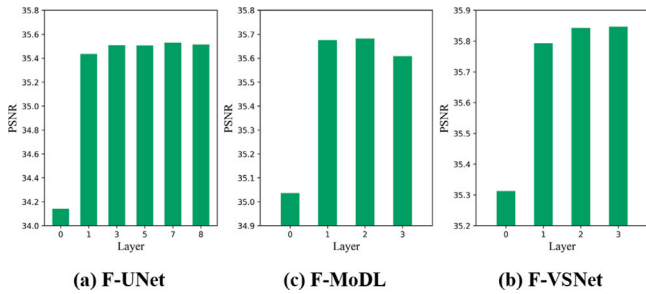


Fig. 7. PSNR results on the knee validation dataset at 8x acceleration when FCB is applied to different layers in the model. (a) Results for UNet. (b) Results for MoDL. (c) Results for VSNet.

and F-MoDL and F-VSNet suppressed the aliasing artifacts to a great extent. The zoomed region and error maps showed that all models with FCB had better performance in reconstructing detailed cartilage and edge information. At 12x acceleration, MoDL and VSNet showed even heavier artifacts, whereas the four FCB models achieved more accurate reconstructions, with F-E2EVar exhibiting slightly smaller residual error.

Example comparison between F-MoDL and other methods is shown in Fig. 6. F-MoDL accurately reconstructed the brain structure and texture, whereas the results of ViT and Ksp exhibited obvious artifacts. ReconFormer and SwinMR produced overly smooth results. Although MDRecon prevented heavy artifacts, it still exhibited more noise-like errors than F-MoDL, as evident in the zoomed-in regions and error maps.

#### 4.6. Analysis of FCB

As a Plug-and-Play block, FCB can be utilized in any convolutional layer of CNN models. Here, the impact of implementing FCB in different layers was further explored. Fig. 7 shows the PSNR results of F-UNet, F-MoDL and F-VSNet in the knee dataset. The x-axis represents the number of convolution layers that were replaced by FCB from the last, with 0 indicating the baseline model without FCB and 1 indicating only the last layer being replaced by FCB. The results for different FCB layers were very close, and using just one FCB layer significantly improved the reconstruction performance. As a single FCB also has a global RF, as referred in Section 3.1, this phenomenon suggests that only replacing the last layer with FCB is capable of incorporating global information into the reconstruction process, aligning with the findings in Rao et al. (2022).

To demonstrate that FCB can scale up the RF while maintaining low computational complexity than traditional convolution, the reconstruction performance and runtime between UNet with different kernel

sizes, and F-UNet were compared. The performance of FasterFC in the same UNet architecture was also compared. As shown in Table 6, enlarging the kernel size of UNet improved the reconstruction metrics, but there was no further improvement when the kernel size was larger than 11. On the other hand, the runtime quickly increased with larger kernel sizes, the runtime of UNet with kernel size of 21 was nearly 3 times as long as that of UNet with standard  $3 \times 3$  convolution kernel. Notably, the runtime of F-UNet was shorter than the best traditional UNet with an  $11 \times 11$  kernel size, aligning with the theoretical operations presented in Table 1. FasterFC performed better than the  $3 \times 3$  convolution and slightly better than the  $7 \times 7$  version, but its runtime was comparable to the  $21 \times 21$  due to the extra point-wise convolution. Given the homogeneity of the Fourier transform, the global connection in FasterFC primarily stems from its Fourier nonlinear layer, which may explain its limited performance. Furthermore, F-UNet achieved the best reconstruction performance, significantly surpassing all the large-kernel UNet variants and FasterFC-UNet, which can be attributed to the global and adaptive receptive field of FCB.

Furthermore, a series of ablation experiments were carried out on UNet-based models using the knee dataset to investigate the effect of all the innovations in the proposed method, including circular padding, depth-wise convolution, shortcut, FCB, and re-parameterization during training. As shown in Table 7, all the modifications contributed to the improved reconstruction performance with high statistic significance. Although circular padding, depth-wise convolution, and shortcut may seem to contribute less, they are actually crucial innovations to embed FCB into CNN. Notably, these modifications have also been employed in the baseline models in Sections 4.4 and 4.5 and did not contribute the over-performance of FCB when comparing.

To explore the layer-wise RF of the proposed FCB, the convolution kernels of UNet and F-UNet were visualized (Fig. 8) by showing the spectrum of the kernel with the maximum norm in each layer. For the baseline UNet, the spectral visualization map appeared very smooth, and the distribution varies across different layers. In contrast, for F-UNet (FCB was deployed in the last 6 layers), the spectral map of FCB kernel exhibited more sharp features and sparsely focused on the zero frequency, which was consistent with the distribution of real k-space data. Furthermore, to illustrate the size in spatial domain of trained FCB kernel, the rank of the trained kernels' spectrum in both models were calculated and showed in Fig. 8 (the white numbers). Due to the orthogonality of Fourier Transform, the rank is consistent between convolution kernel and its spectrum (Proakis, 2007), so the rank of spectrum is a lower bound of kernel's size in spatial domain, as described in Eq. (6). The rank of FCB kernels could reach high values, such as 143, while the ranks of normal convolutions are typically lower than 3. In terms of layer-wise dimension, it is evident that FCB significantly scaled up the size of convolution and, namely, the RF.

**Table 6**

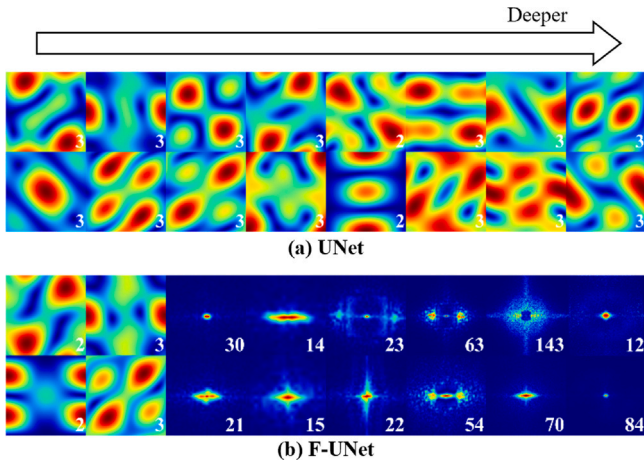
Comparison the reconstruction performance and runtime between UNet with different Kernel sizes, FasterFC-UNet and F-UNet on the knee dataset at 8× acceleration.

|           | k = 3           | k = 7           | k = 11          | k = 15          | k = 21          | FasterFC        | FCB             |
|-----------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|
| PSNR      | 34.23 ± 2.15    | 34.40 ± 2.16    | 34.79 ± 2.23    | 34.83 ± 2.23    | 34.42 ± 2.19    | 34.44 ± 2.08    | 35.52 ± 2.34    |
| p         | ***             | ***             | ***             | ***             | ***             | ***             | —               |
| SSIM      | 0.8955 ± 0.0338 | 0.8981 ± 0.0336 | 0.9030 ± 0.0329 | 0.9037 ± 0.0328 | 0.8984 ± 0.0337 | 0.9020 ± 0.0328 | 0.9136 ± 0.0321 |
| p         | ***             | ***             | ***             | ***             | ***             | ***             | —               |
| Time (ms) | 6.88            | 8.58            | 10.87           | 14.3            | 21.74           | 21.61           | 10.09           |

**Table 7**

Ablation study assessing the impact of modifications and the Re-parametrization method on the knee validation dataset at 8× acceleration.

| Circular | Depth-wise | ShortCut | FCB | Re-Para | PSNR         | p   | SSIM            | p   |
|----------|------------|----------|-----|---------|--------------|-----|-----------------|-----|
| ✓        |            |          |     |         | 33.38 ± 2.00 | *** | 0.8802 ± 0.0345 | *** |
| ✓        |            |          |     |         | 33.57 ± 2.00 | *** | 0.8858 ± 0.0338 | *** |
| ✓        | ✓          |          |     |         | 33.82 ± 2.13 | *** | 0.8892 ± 0.0347 | *** |
| ✓        | ✓          | ✓        |     |         | 34.23 ± 2.15 | *** | 0.8955 ± 0.0338 | *** |
| ✓        | ✓          | ✓        | ✓   |         | 35.22 ± 2.30 | *** | 0.9092 ± 0.0330 | *** |
| ✓        | ✓          | ✓        | ✓   | ✓       | 35.52 ± 2.34 | —   | 0.9136 ± 0.0321 | —   |

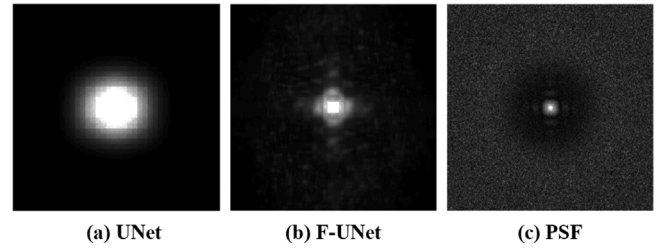


**Fig. 8.** Spectral visualization of convolution kernels in UNet and F-UNet (FCB was deployed in the last 6 layers). (a) Spectral maps of UNet. (b) Spectral maps of F-UNet. The spectral amplitude of convolution kernels or the kernels in FCB are shown from the left to right along the depth of UNet. The upper and lower rows correspond to the double convolution layers in each UNet block. The ranks of spectrum of kernels are noted in the bottom right-hand corner of each map.

The effective receptive field (ERF) (Luo et al., 2016) was also calculated to investigate the whole RF of FCB enhanced CNN, which can be evaluated through back-propagation techniques:

$$\text{ERF} = \sum_k \sum_{i,j} \frac{\partial y_{\text{cen}}^k}{\partial x_{i,j}^k} \quad (10)$$

The ERF is an average value among the dataset dimension  $k$ , where  $y_{\text{cen}}^k$  represents the center activations in the final output layer of the CNN and  $x_{i,j}^k$  represents the  $i$ th row and  $j$ th column element of the input image of the CNN. Fig. 9 displays the ERFs of UNet and F-UNet. Because the final output layer of UNet has a large size than common classification networks,  $y_{\text{cen}}^k$  includes the center pixels of a  $10 \times 10$  region chosen for back-propagation. The ERF of UNet exhibited a distribution similar to a normal distribution with a mean centered at the image center. On the other hand, the ERF of F-UNet not only focused on the image center, but also covered a larger region. As a comparison, the point spread function (PSF) of the 2D sampling-pattern of the k-space used in our experiments is shown (Fig. 9(c)). It can be observed that the ERF of F-UNet covered a wider range and was closer to the PSF of k-space sampling in MR imaging. This is more conducive for the network to utilize the aliased signal and recover the original image.



**Fig. 9.** (a) ERF of UNet. (b) ERF of F-UNet. (c) PSF of the 2D Poisson sampling pattern. The ERF of F-UNet covered a larger region, closing to the sampling PSF.

## 5. Conclusion

This paper introduced a novel convolution block design with a global receptive field for MRI reconstruction CNNs. The experimental results showed that the proposed FCB effectively improved the reconstruction performance of the baseline CNN models. At different undersampling rates, models enhanced with FCB achieved better quantitative metrics in various datasets, even with additional noise added. Furthermore, these models exhibited superior capability in recovering intricate details, including texture and edges. Notably, the models with FCB outperformed Vision Transformers, which are considered mainstream models with large receptive fields. Additionally, FCB models also surpassed methods that embed k-space data to enhance long-range connections. FCB also demonstrated low computational complexity, with experiments showing that its runtime is significantly less than Vision Transformers and comparable to traditional CNNs with an  $11 \times 11$  convolution kernel.

Through visualization, it is demonstrated that the proposed FCB effectively scales up the RF of CNNs. Unlike other approaches that focus on the architecture design of CNNs, our approach pays attention to the basic convolution layer in CNNs. Some other works use cascaded or pyramidal architectures (Schlemper et al., 2017; Chen et al., 2022; Sriram et al., 2020a) to capture long-distance correlation. Our proposed FCB serves as a Plug-and-Play block that has the potential to be incorporated into these various CNN architectures. Moreover, FCB processes data in the hidden layer as real values and utilizes conjugate symmetry in the frequency domain to save memory and time. For methods building complex CNNs (Wang et al., 2020; Cole et al., 2021), FCB could also be smoothly integrated in complex mode.

The proposed FCB approach still has some limitations. Although its usage memory is comparable to spatial convolution, the number of parameters in FCB is still significantly higher, leading to increased storage memory costs. This presents a tradeoff between the memory

and speed when a large RF is desired. Additionally, the proposed FCB involves repeated FFTs and IFFTs, which may impact computing efficiency. Some works (Ayat et al., 2019; Watanabe and Wolf, 2021) have attempted to design pure spectral-based CNNs to address this issue, but the question of activation functions in the Fourier domain remains open. Future research in this area could explore ways to improve the computing efficiency of FCB.

## CRedit authorship contribution statement

**Haozhong Sun:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Yuze Li:** Resources, Methodology, Formal analysis, Conceptualization. **Zhongsen Li:** Writing – review & editing, Methodology, Conceptualization. **Runyu Yang:** Validation, Methodology. **Ziming Xu:** Writing – review & editing, Visualization. **Jiaqi Dou:** Software, Methodology. **Haikun Qi:** Writing – review & editing. **Huijun Chen:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Project administration, Methodology, Investigation, Funding acquisition, Conceptualization.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Data availability

Data will be made available on request.

## Declaration of Generative AI and AI-assisted technologies in the writing process

Statement: During the preparation of this work the authors used OpenAI's tool ChatGPT in order to improve readability and language in revision process. After using this tool, authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

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